

Sparse Arrays for Near-Field Cell-Free XL-MIMO

A plain-language guide, literature map, and plan for extending the ICASSP and VTC papers into a Transactions paper

(Version 2: every claim checked against the full texts of the cited papers)

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How to read this document

Sections 2–3 explain what a sparse array is and why it matters in the near field, with numbers computed for *your* system ($L = 8$ APs, $N = 64$ antennas, $f_c = 3$ GHz, $300 \times 300 \text{ m}^2$). Section 4 lists the rules to respect, Section 5 collects the equations, Section 6 maps the literature and Section 7 states the gap. Sections 8–10 give options and a plan for the Transactions paper.

Citation conventions and verification status

- “[x, eq. (y)]”, “[x, Lemma z]” etc. point to the exact place in the cited paper. Equation, lemma and section numbers refer to the arXiv versions listed in the bibliography.
- “**Derived here**” marks a result that does not appear in a paper. These are short derivations under the stated approximation; check them before using them in a paper.
- “**Computed here**” marks numbers obtained with the exact NUSW model for your parameters.
- The full texts of [5–9, 11–14] were read and checked. All other references were checked only at the level of title, authors and abstract; this is stated in Table 4 at the end.

Contents

1	Where your work stands, and what a Transactions paper needs	3
2	What is a sparse array?	3
3	Why sparse arrays matter in the near field: numbers for your system	4
3.1	The near-field region grows with η^2	4
3.2	Range focusing needs the user well inside R	4
3.3	More degrees of freedom	5
3.4	Grating lobes: the price, and what the near field changes	5
4	Rules and conditions to respect	6
5	Mathematical background	6
5.1	Channel model with a sparse AP	6
5.2	Fresnel form, surrogate distance and array gain	7
5.3	Grating lobes in the near field	7
5.4	Inter-user correlation and resolvability	7
5.5	Effect on your cross-AP information surplus	7
6	Literature map	8

7	The gap	9
8	Options for bringing sparse arrays into your model	10
9	Channel estimation with sparse APs (connects to your VTC paper)	10
10	A plan for the Transactions paper	11
	Verification status of the references	11

1 Where your work stands, and what a Transactions paper needs

ICASSP paper. Cross-AP list-based SIC for hybrid near-/far-field cell-free XL-MIMO with *perfect* CSI [1]: the candidate list is formed from both the cluster-fused and the all-AP fused soft estimates, cancellation is network-wide, and an LDPC-coded IDD scheme is added.

VTC paper. The same detector with *estimated* CSI from polar-domain channel estimation over all AP-UE pairs [2, 3].

What a Transactions paper needs. A new technical dimension (not only a merge of the two conference papers), analysis, and a broad numerical study. Sparse APs are a strong candidate, because they change three things your papers rely on:

1. *where* the near field is: the Rayleigh distance grows with the square of the aperture [4, 5];
2. *how well* co-angular users can be separated in range: resolution improves with the aperture [7–9];
3. *what goes wrong*: grating lobes [6–8]. Your cross-AP fusion is a natural way to fix this (Section 3).

2 What is a sparse array?

A conventional uniform linear array (ULA) has spacing $d = \lambda/2$. A **sparse array** uses spacing larger than $\lambda/2$, or non-uniform positions [7, 8, 10].

Main families.

- **Uniform sparse ULA** (“linear sparse array” (LSA) in [8], “uniform sparse array” (USA) in [7]). Spacing $d = \eta \lambda/2$ with integer sparsity factor $\eta > 1$; [8] writes U and [7] writes p for this factor. Positions:

$$p_n = \left(n - \frac{N+1}{2}\right) \eta \frac{\lambda}{2}, \quad n = 1, \dots, N. \quad (1)$$

- **Thinned array**: a dense array in which only a subset of elements is active. The aperture stays the same with fewer RF chains; the active set can be reconfigured [11].
- **Non-uniform sparse arrays**: minimum-redundancy [22], nested [23] and co-prime [24] arrays, the extended co-prime array (ECA) of [8], and position-optimised arrays [7]. Non-uniform spacing removes the exact periodicity that causes grating lobes [7, 8].
- **Modular arrays**: dense sub-arrays separated by large gaps [12, 13].

Two ways to compare with a dense array. These give different conclusions, so always state which one you use:

- **Same number of antennas N** : the sparse array has a larger aperture. This is the setting of [8, 9] (“given the same number of array elements”).
- **Same aperture**: about N/η antennas, which is the motivation of thinning [11] and of modular arrays with a limited number of antennas [13].

3 Why sparse arrays matter in the near field: numbers for your system

3.1 The near-field region grows with η^2

With aperture $D = (N - 1)d$, the Rayleigh distance used in your papers (eq. (2) of [2], following [4]) is $R = 2D^2/\lambda$. With $d = \eta\lambda/2$:

$$R(\eta) = \frac{(N - 1)^2 \eta^2 \lambda}{2} = \eta^2 R(1) \quad (\text{derived here from } R = 2D^2/\lambda). \quad (2)$$

Doubling the spacing makes the near-field region four times larger.

3.2 Range focusing needs the user well inside R

Being inside R is not enough for a sharp focus in distance.

- For a *planar* array focusing on its axis, Björnson *et al.* prove that the 3-dB depth of focus is finite only if the focal distance $F < d_{\text{FA}}/10$, where $d_{\text{FA}} = 2D_{\text{diag}}^2/\lambda$ uses the array diagonal [5, eq. (17), Theorem 1, eq. (23)].
- For a *linear* (dense or sparse) array, Zhou *et al.* give the same structure [8, Proposition 2, eq. (22)]:

$$\text{BD}_{3\text{dB}} = \begin{cases} \frac{2r_0^2 r_{\text{th}}}{r_{\text{th}}^2 - r_0^2}, & r_0 < r_{\text{th}}, \\ \infty, & r_0 \geq r_{\text{th}}, \end{cases} \quad r_{\text{th}} \approx \frac{Q^2 U^2 d_0 \cos^2 \theta_0}{4\beta_{3\text{dB}}^2}, \quad \beta_{3\text{dB}} = 1.6, \quad (3)$$

with Q antennas, sparsity U (our η), $d_0 = \lambda/2$ and user angle θ_0 .

Since $R \approx Q^2 U^2 d_0$ for a ULA with $d_0 = \lambda/2$, (3) gives (derived here)

$$r_{\text{th}} \approx \frac{R \cos^2 \theta_0}{10.24}, \quad (4)$$

which matches the $d_{\text{FA}}/10$ rule of [5], and adds the factor $\cos^2 \theta_0$: focusing weakens towards end-fire and disappears at $\theta_0 = \pm 90^\circ$ [8, Remark 1]. Zhou *et al.* also note that a higher sparsity gives a more prominent focusing effect [8, Remark 1].

Table 1: Numbers computed here for your system ($f_c = 3$ GHz, $\lambda = 0.1$ m), with the exact NUSW model. r_{th} is (4) at broadside ($\theta_0 = 0$). “Corr.” is $|\mathbf{b}_1^H \mathbf{b}_2|$ between the unit-norm array vectors of two users at the *same* angle and different ranges; small values mean the AP can separate them by range.

η	N	spacing [m]	aperture D [m]	R [m]	r_{th} [m]	Corr. 30 vs 120 m ($\sin \theta = 0$)	Corr. 40 vs 80 m ($\sin \theta = 0$)
1	64	0.05	3.15	198	19.4	0.83	0.96
2	64	0.10	6.30	794	77.5	0.36	0.46
4	64	0.20	12.6	3175	310	0.15	0.24
2	32	0.10	3.10	192	18.8	0.83	0.96
4	16	0.20	3.00	180	17.6	0.83	0.96

What Table 1 tells you about your current papers

- Each user in your simulations is placed 20–150 m from its anchor AP (other APs are farther). With compact APs ($\eta = 1$), $r_{\text{th}} \approx 19$ m at broadside, so **almost no link has a finite beam depth**. The links are near-field by the Rayleigh criterion of [2], but two co-angular users at 30 m and 120 m still have correlation 0.83.
- With the *same 64 antennas* at $\eta = 2$ the correlation drops to 0.36, and at $\eta = 4$ to 0.15. This agrees with the finding that larger spacing improves spatial resolution and multi-user sum rate [7, 9].
- With the *same aperture* (rows 4–5) the correlation is unchanged: range resolution depends on the aperture, not on the number of antennas. A thinned array gives the same separation with $2\times$ or $4\times$ fewer antennas, which is the motivation of [11].

3.3 More degrees of freedom

Wang *et al.* define the effective degrees of freedom (EDoF) of a channel $\mathbf{H}$ as [9, eq. (13)]

$$\text{EDoF} = \left(\frac{\text{tr}(\mathbf{H}\mathbf{H}^H)}{\|\mathbf{H}\mathbf{H}^H\|_F} \right)^2, \quad (5)$$

and prove, for single-user near-field sparse MIMO, a closed-form EDoF that increases with the array sparsity until it reaches $\min(N_t, N_r)$ [9]. Chen *et al.* give a constraint under which the EDoF of sparse UPAs can be estimated in closed form [6]. For multi-user detection, the analogous effect is a better-conditioned local channel matrix $\mathbf{H}_l$, i.e. the rank enhancement of eq. (10) in [2].

3.4 Grating lobes: the price, and what the near field changes

A uniform sparse ULA repeats its beam in $\sin \theta$ with period $2/\eta$. Its grating lobes are at [8, Lemma 5, eq. (23)]

$$\sin \theta_u = \sin \theta_0 + \frac{2u}{\eta}, \quad |\sin \theta_u| \leq 1, \quad (6)$$

and there are $2(\eta - 1)$ of them over the whole angular range, each with beam height 1 (as strong as the main lobe) [8, Lemma 5].

- **Far field:** the grating lobes have the same peak power as the main lobe, so there is no suppression [6, Sec. IV, after Corollary 3].
- **Near field, at the focal distance:** the grating lobes are suppressed, and more so for shorter focal distance and larger spacing [6, Corollary 3, eq. (40)]. The same authors warn that “particular attention should be paid to the few grating lobes that retain relatively high power” [6, Sec. IV]. The near-field model also suppresses the grating lobes of modular arrays better than the far-field model [12].
- **Near field, at a shifted range:** the grating lobe is *refocused* at another distance, where it is again as strong as the main lobe. Zhou *et al.* show that at the grating-lobe angle the beam pattern depends on $\cos^2 \theta_0 / r_0 - \cos^2 \theta_u / r$ [8, Lemma 6, eq. (24)], which vanishes at

$$r' = r_0 \frac{\cos^2 \theta_u}{\cos^2 \theta_0} = r_0 \frac{1 - \sin^2 \theta_u}{1 - \sin^2 \theta_0} \quad (7)$$

(rewritten here from [8, eq. (24)]). Zhou *et al.* describe these lobes as “focused on specific regions” with power comparable to the main lobe [8, Abstract, Remark 2].

Example computed here (your array, $\eta = 2$, focus at $\sin \theta_0 = 0.3$, $r_0 = 40$ m): gain 0.38 at the grating-lobe angle $\sin \theta_u = -0.7$ at the same range; gain 0.89 at the same angle at $r' = 22.4$ m from (7); gain 1.00 for a far-field user at that angle.

Why this is good news for a cell-free paper (proposed here)

In a hybrid-field cell-free network each AP sees a user at a different angle and range, so the ambiguity point (7) of a user is different at every AP. Two users that collide on a grating lobe at one AP are unlikely to collide at all APs. A detector that fuses *all* APs, like your cross-AP list SIC [1, 2], can then resolve what one AP, or the serving cluster, cannot. Sparse APs should therefore make the information surplus Δ_k of your papers (eq. (26) of [2]) larger. This is a hypothesis to prove; no paper found in this search has studied it.

4 Rules and conditions to respect

- R1. State the comparison basis:** same N [8, 9] or same aperture [11, 13].
- R2. Recompute the near/far-field boundary.** $R \propto \eta^2$ (2). In your $300 \times 300 \text{ m}^2$ area, $\eta = 2$ gives $R = 794 \text{ m}$, so almost every link becomes near-field and the *hybrid* regime disappears (computed here). Keep a hybrid network with a larger area, with mixed APs, or with a boundary that depends on angle as well [4, 21].
- R3. Range focusing needs** $r_0 < r_{\text{th}}$ (3)–(4) [5, 8]. Report the fraction of links inside r_{th} , and remember the $\cos^2 \theta_0$ factor.
- R4. Account for grating lobes everywhere:** inter-user correlation, clustering, and any use of estimated angles. Far-field links of a sparse AP see grating lobes at full strength [6].
- R5. Estimation dictionary.** The angular grid must be about η times finer, because the main lobe narrows to a null-to-null width of $4/(N\eta)$ in $\sin \theta$ [8, Proposition 1]. For one AP, the channel vector is periodic in the surrogate-distance/angle domain with period $2/\eta$, so a dictionary over one period is enough to estimate the *vector* [7, Sec. IV-A, eq. (47)]; see Section 9 for what this means for angle and range.
- R6. Keep the element model consistent.** In your code the far-field correlation matrix uses the spacing in wavelengths ($d_H = 0.5$); it must become $\eta/2$ together with the NUSW positions (code check, done here).
- R7. Hardware and channel assumptions.** A larger aperture raises cost, size and spatial non-stationarity concerns; thinning keeps the hardware but switches elements [11]. Wideband operation makes grating lobes move with frequency [10]; your system is narrowband, so state it. Recent near-field cell-free work uses dense planar arrays with LoS and NLoS components [14]; your model is a ULA with LoS (near-field) and correlated Rayleigh (far-field) links, so state the difference.
- R8. Non-uniform arrays** (ECA, position-optimised, nested, co-prime) reduce the grating-lobe peaks: the ECA “can greatly suppress the beam power of near-field grating-lobes” but has more of them [8, Abstract, Sec. III], and optimised non-uniform arrays have no grating lobes in the SD-A domain [7, Sec. IV-A].

5 Mathematical background

5.1 Channel model with a sparse AP

Keep the NUSW model of your papers (eq. (6) of [2], from [4]) and change only the positions (1):

$$[\mathbf{h}_{l,k}^{\text{NF}}]_n = \sqrt{\beta_{l,k}} \frac{d_{l,k}}{d_{n,l,k}} e^{-j \frac{2\pi}{\lambda} d_{n,l,k}}, \quad d_{n,l,k} = \sqrt{d_{l,k}^2 + p_n^2 - 2 d_{l,k} p_n \sin \theta_{l,k}}. \quad (8)$$

The far-field model becomes $\sqrt{\beta_{l,k}} \mathbf{a}(\theta_{l,k})$ with $[\mathbf{a}(\theta)]_n = e^{j \frac{2\pi}{\lambda} p_n \sin \theta}$ (eqs. (8)–(9) of [2] with the new p_n).

5.2 Fresnel form, surrogate distance and array gain

The second-order expansion $d_n \approx r - p_n s + \frac{p_n^2(1-s^2)}{2r}$, $s = \sin \theta$, is eq. (7) of [2] and the basis of the polar-domain model of [3]. Chen *et al.* call

$$b = \frac{1 - s^2}{2r} \quad (9)$$

the *surrogate distance* [7, eq. (6)]. With it, the unit-norm steering vector is

$$[\mathbf{b}(s, b)]_n \approx \frac{1}{\sqrt{N}} \exp\left(j \frac{2\pi}{\lambda} (p_n s - p_n^2 b)\right), \quad (10)$$

and the normalized gain when focusing at (s_0, b_0) and looking at (s, b) is (derived here from (10))

$$G(s, b) = \left| \frac{1}{N} \sum_{n=1}^N \exp\left(j \frac{2\pi}{\lambda} [p_n(s - s_0) - p_n^2(b - b_0)]\right) \right|^2. \quad (11)$$

5.3 Grating lobes in the near field

With (1), the linear phase in (11) is a multiple of 2π for every n when $s - s_0 = 2u/\eta$, which gives (6) [8, Lemma 5]. At those angles only the quadratic term remains:

1. If $b = b_0$, the quadratic phase vanishes and the gain is 1. In physical coordinates this is (7) [8, Lemma 6]. In the (s, b) domain, the focus pattern is therefore exactly periodic with period $2/\eta$ [7, Sec. IV-A].
2. At the same physical range $r = r_0$, $b - b_0 = (s_0^2 - s_u^2)/(2r_0)$. The largest quadratic phase, at the array edge $p = D/2$, is (derived here)

$$\Phi_{\max} = \frac{\pi R}{8 r_0} |s_u^2 - s_0^2|, \quad (12)$$

so the lobe is strongly attenuated when $r_0 \lesssim R|s_u^2 - s_0^2|/8$. Since $R \propto \eta^2$, sparser arrays and closer users give more suppression, consistent with [6, Corollary 3].

3. As $r_0 \rightarrow \infty$, $b_0 \rightarrow 0$ and $\Phi_{\max} \rightarrow 0$: no suppression in the far field [6].

5.4 Inter-user correlation and resolvability

At AP l define $\mu_l(i, j) = |\mathbf{h}_{l,i}^H \mathbf{h}_{l,j}| / (\|\mathbf{h}_{l,i}\| \|\mathbf{h}_{l,j}\|)$. Users i, j are unresolvable at AP l when $\mu_l(i, j) \approx 1$, for example when one sits at the other's ambiguity point (7). A network-level measure for a set of APs $\mathcal{T}$ is (proposed here)

$$\mu_{\text{net}}(i, j; \mathcal{T}) = \min_{l \in \mathcal{T}} \mu_l(i, j). \quad (13)$$

The difference between $\mathcal{T} = \mathcal{S}_k$ (cluster) and $\mathcal{T} = \{1, \dots, L\}$ is what cross-AP detection exploits. For a single uniform sparse array, Chen *et al.* show that the total coverage of the main lobes (including grating lobes) is still no larger than that of a dense half-wavelength ULA with the same number of antennas, so the sparse array has higher spatial resolution despite grating lobes [7, Property 3].

5.5 Effect on your cross-AP information surplus

Your papers define the per-AP SINR $\gamma_{l,k}$ (eq. (23) of [2]) and the surplus (eq. (26) of [2])

$$\Delta_k \approx \frac{\sum_{l \notin \mathcal{S}_k} \gamma_{l,k}}{(1 + \sum_{l \in \mathcal{S}_k} \gamma_{l,k})(1 + \sum_{l=1}^L \gamma_{l,k})}. \quad (14)$$

With sparse APs, $\gamma_{l,k}$ collapses at APs where user k sits on an ambiguity point of a strong interferer and stays large elsewhere. When such an AP is in the cluster, Δ_k grows (hypothesis proposed here). A bound on $\mathbb{E}[\Delta_k]$ as a function of η would be a Transactions-level result.

6 Literature map

Table 2: Key works, checked against their full texts unless stated otherwise in Table 4.

Work	What they did	What is missing for you
<i>Sparse arrays in near-field communications (single array)</i>		
Chen, Ren, Pan, Wang, Wang [6]	Sparse UPAs, Green’s-function channel model. Closed-form beam power vs distance and angle, lobe length and grating-lobe suppression ratio: distance focusing and grating-lobe suppression both improve as the focal distance decreases or the spacing increases (Corollaries 1–3). Closed-form EDoF under a parameter constraint.	Single array; no cell-free; no detection or estimation.
Chen, Qi, Li, Dobre [7]	Uniform and non-uniform sparse arrays for near-field multi-user communication. Larger spacing improves spatial resolution and sum rate (Property 3); SCA-based antenna-position optimisation for non-uniform arrays; channels sparse in the surrogate distance–angle (SD-A) domain, with on-grid SDA-OMP and off-grid SDA-ISRCE estimators.	Single array; precoding and estimation only; no SIC or list detection.
Zhou, You, Zhang, Chen, Shi [8]	Near-field beam pattern of the LSA: main-lobe width and depth (Propositions 1–2), $2(U - 1)$ grating lobes of height 1 (Lemma 5), focused at specific ranges (Lemma 6, Proposition 3). ECA with smaller grating-lobe power but more lobes; hybrid beamforming for both.	Downlink, single array.
Wang, Feng, Zeng, Jin, Yuen, Clerckx, Zhang [9]	Sparse MIMO with the same number of elements: narrower main lobe, larger near-field region; closed-form near-field EDoF increasing with sparsity up to $\min(N_t, N_r)$; multi-user interference suppression in near and far field.	No cell-free, no estimation.
Hussain, Abdallah, Celik, Björnson, Eltawil [11]	Reconfigurable array thinning (particle swarm optimisation) for near-field MU-MIMO; performance comparable to movable antennas without moving parts.	Single array.
Li, Min, Zeng, Jin, Dai, Yuan, Zhang [10]	Survey of sparse MIMO for ISAC: co-arrays, grating-lobe suppression, code-books.	Sensing focus.
<i>Modular arrays</i>		

Work	What they did	What is missing for you
Li, Dong, Zeng, Jin, Zhang [12]	Modular XL-array: sub-array USW models; better resolution than a collocated array with the same elements but grating lobes; the near-field model suppresses grating lobes better than the far-field UPW model.	Single modular array.
Kosasih, Björnson [13]	Demir, Modular linear arrays: beamwidth and beamdepth analysis; ideal focusing by choosing the number of antennas per ULA; localization by fusing per-ULA estimates through least-squares triangulation; parametric channel estimation.	Not cell-free; the per-ULA fusion idea is close to your cross-AP fusion.
<i>Near-field cell-free</i>		
Wang, Zhang, Xu, Niyato, Ai, Mao, Han [14]	Uplink SE framework for near-field cell-free XL-MIMO with <i>dense planar arrays</i> (UPAs) and LoS+NLoS channels; centralized and local MMSE combining for <i>arbitrary</i> channel estimators; five low-complexity combiners (matrix approximation, SSOR).	Dense arrays; linear combining only; no list/SIC detection.
Lei <i>et al.</i> [15]	Distributed activity detection in hybrid near-far field cell-free.	Activity detection, not data detection.
Lei, Wang, Xiao, Zhang, Ai [16]	Uplink performance of cell-free XL-MIMO.	Dense arrays.
Mylonopoulos, Interdonato, Buzzi, Liu [17]	Near-field beamfocusing in cell-free networks using LoS from distributed APs.	Downlink focusing.
Xie <i>et al.</i> [18]	Beam focusing for near-field cell-free massive MIMO with few antennas per AP; includes a sparse-array angle-based focusing scheme (abstract level).	Downlink focusing; no detection, no estimation.
<i>Foundations</i>		
Cui, Dai [3]; Wei, Dai [19]	Polar-domain dictionary and OMP; hybrid-field estimation.	Dense arrays.
Björnson, Demir, Sanguinetti [5]	Fraunhofer array distance and finite beam depth only below $d_{FA}/10$ (planar array).	—
Lu, Zeng <i>et al.</i> [4]; Wang <i>et al.</i> [20]	Tutorials on near-field XL-MIMO.	—
Moffet [22]; Pal, Vaidyanathan [23, 24]	Minimum-redundancy, nested and co-prime arrays.	Classic array processing.

7 The gap

Within the limits of this search:

1. Sparse arrays in the near field are studied for **single arrays**: beam patterns, EDoF, precoding, thinning, estimation and sensing [6–13].
2. Near-field **cell-free** work uses dense arrays and studies combining, activity detection or downlink focusing [14–17]. Only [18] mentions sparse arrays, for downlink beam focusing.
3. **No work was found** on uplink data detection (SIC, list detection, IDD) or channel estimation for *hybrid-field cell-free* networks with *sparse APs*.

4. **No work was found** that uses *multi-AP fusion* to resolve per-AP grating-lobe ambiguities. Kosasih *et al.* fuse sub-array estimates within one modular array [13], which is the closest idea.

Point 4 is the strongest claim, because it connects directly to the cross-AP idea of your papers. “Not found” means “not found by this search”.

8 Options for bringing sparse arrays into your model

Option	What changes	Pros and cons
A. Uniform sparse ULA, same N	Spacing $\eta\lambda/2$ at every AP; $R \propto \eta^2$; better range resolution (Table 1) [8, 9].	+ Biggest gain in separation. – Larger AP; hybrid regime may vanish (R2); grating lobes.
B. Thinned, same aperture	About N/η active antennas over the same aperture [11].	+ Same R and hybrid regime; cheaper local combiners (an $N \times N$ inverse costs $\mathcal{O}(N^3)$, computed here). – Grating lobes with no resolution gain; lower array gain.
C. Non-uniform (ECA, optimised, nested, co-prime)	Positions from [7, 8, 22–24].	+ Lower grating-lobe peaks [7, 8]. – Harder analysis.
D. Modular AP	Dense sub-arrays with gaps [12, 13].	+ Practical, known theory. – Grating lobes at sub-array level.
E. Heterogeneous APs	Some APs dense, some sparse.	+ Keeps a hybrid-field network. – Larger design space.
F. Network as a sparse array	Treat the L APs as one distributed array.	+ What cross-AP fusion already exploits. – Synchronisation assumptions.

Recommendation

Use **Option A** as the main study ($\eta \in \{1, 2, 4\}$, same N) and **Option B** as the fairness check (same aperture). Add **Option E** for a design-insight figure if space allows; Option C is a good follow-up.

9 Channel estimation with sparse APs (connects to your VTC paper)

Three points, the second one corrected in this version:

1. **Finer angular grid.** Use about η times more angle samples, because the main lobe is η times narrower [8, Proposition 1]. Sample the range in the surrogate distance (9) [7] or in $1/r$ per angle [3].
2. **Periodicity: harmless for the channel vector, harmful for parameters.** For one uniform sparse AP, $\mathbf{b}(s, b)$ and $\mathbf{b}(s + 2/\eta, b)$ are nearly identical (10). Chen *et al.* therefore sample the angle over a single period $[-1/p, 1/p]$ [7, eq. (47)]: the estimated *channel vector* is correct even if the angle is aliased. The aliasing only matters when the estimated angle and range are *used*, for example for localization, to combine information across APs, or in a clustering rule based on geometry. (The authentic estimator in your MATLAB code resolves a related end-fire alias for $\eta = 1$, where $s = -1$ and $s = +1$ coincide.)

3. **Cross-AP location-domain refinement (proposed here).** A user has one position $\mathbf{u}_k$, which fixes $(\theta_{l,k}, d_{l,k})$ at every AP. Estimating $\mathbf{u}_k$ jointly from all near-field APs removes the per-AP aliases, because an alias consistent with AP l is not consistent with AP l' . This extends to cell-free the per-sub-array fusion by least-squares triangulation of [13]. Off-grid refinement can follow [3, 7].

10 A plan for the Transactions paper

Working title. *Sparse-Array Near-Field Cell-Free XL-MIMO: Grating-Lobe-Aware Channel Estimation and Cross-AP List Detection with Iterative Decoding.*

Contributions (draft).

- C1.** Hybrid-field cell-free model with sparse APs; fraction of near-field links (2) and of focusing links (4) vs. η and area.
- C2.** Grating-lobe analysis for cell-free: per-AP ambiguity points (7) [8], and the probability that a pair is unresolvable at one AP, in a cluster and at all APs.
- C3.** Location-domain cross-AP estimation (Section 9), with NMSE analysis.
- C4.** Cross-AP list SIC and IDD [1, 2], with Δ_k analysed vs. η .
- C5.** Clustering that accounts for grating-lobe interference (separate clustering document).

Figures to plan. (1) fraction of near-field and focusing links vs. η ; (2) one AP's beam pattern for $\eta = 1, 2, 4$ (main lobe, grating lobe at the same range, ambiguity point); (3) NMSE vs. SNR: per-AP OMP, location-domain estimator; (4) BER vs. SNR for the four detectors, $\eta = 1, 2, 4$, perfect and estimated CSI; (5) same- N vs. same-aperture; (6) CDF of Δ_k vs. η , gate firing rate and fronthaul; (7) IDD BER at iteration 3.

Code changes in your MATLAB script.

1. Add `eta`; set `d_ant = eta*lambda/2` and `d_H = eta/2`. Keep N (Option A) or set $N = \text{round}((N_0-1)/\text{eta})+1$ (Option B).
2. `d_Ray` and the near-field mask then update automatically; print the near-field fraction and the fraction with $r < r_{\text{th}}$.
3. Build the polar dictionary with `pd_os_ang` scaled by `eta`. One angular period is enough for the channel vector [7]; if the estimated angle is used elsewhere, add the alias candidates $s \pm 2m/\eta$ to the refinement starts.
4. Apply the same spacing to the centralized-BS benchmark, or state that it stays dense.
5. Keep `genie_csi = false` so that every estimated-CSI result is authentic.

Risks. With $\eta \geq 2$ in $300 \times 300 \text{ m}^2$ nearly all links are near-field; a larger aperture raises spatial non-stationarity, which your model ignores; sparse arrays help range separation only for links inside r_{th} .

Verification status of the references

References

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Table 4: What was checked for each reference.

References	Checked against
[5–9, 11–14]	Full text (statements and equation numbers).
[1, 2]	Your submitted papers.
[3, 4, 25]	As cited in your papers.
[10, 15–24]	Title, authors and abstract only (web search). Read before citing specific results.

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